

CHAPTER 1

INTRODUCTION

In today's rapidly growing digital world, online ordering systems have become an important part of modern businesses. Industries such as restaurants, retail stores, supermarkets, food delivery services, and e-commerce platforms increasingly depend on web-based applications to manage customer orders efficiently. Customers now prefer online ordering platforms because they provide convenience, faster service, easy accessibility, and improved user experience. Instead of visiting stores physically or placing orders manually through phone calls, users can quickly submit their orders through a web application from anywhere at any time.

Traditional order management methods such as handwritten records, spreadsheets, paper bills, and messaging applications are no longer sufficient for handling large volumes of customer orders. These manual approaches are time-consuming, difficult to maintain, and highly prone to human errors. Problems such as duplicate order entries, missing customer details, delayed processing, incorrect order tracking, and inefficient record management are commonly observed in manual systems. As businesses grow, maintaining customer orders manually becomes increasingly complex and affects the overall efficiency and productivity of the organization.

Motivation

Many businesses still rely on manual order management methods such as handwritten records, spreadsheets, or simple messaging systems. These approaches create several problems:

- Duplicate order records
- Difficulty in tracking customer orders
- Delayed order processing
- Lack of proper customer feedback analysis

In the modern digital age, businesses are increasingly adopting online systems to improve their services and operational efficiency. However, many small and medium-scale organizations still depend on traditional order management methods such as handwritten records, spreadsheets, phone calls, and messaging applications. Although these methods are simple to use, they create several challenges when the number of customers and orders increases.

Existing System

Existing systems mainly depend on:

- Manual order booking
- Phone call-based ordering
- Spreadsheet management
- Basic web forms without database integration

The existing order management systems used in many businesses are mostly traditional and semi-digital methods. Small shops, restaurants, and service providers often maintain customer orders manually in notebooks or registers.

Proposed System

The proposed system automates the entire ordering process using modern Java web technologies. The system is designed as a web-based application that allows customers to place orders easily through an online form. All order-related activities such as order submission, order ID generation, data storage, feedback analysis, and confirmation message display are performed automatically by the system.

The application uses Java Servlets for request processing, JDBC for database connectivity, Java Collections for managing order records, and String Processing techniques for feedback analysis and order ID generation.

Features include:

- Customer order submission through web form
- Dynamic order ID generation
- JDBC database storage
- Feedback sentiment checking using string operations

The proposed Online Order Management System is developed to overcome the limitations of traditional manual order handling methods. Customers can place their orders through a user-friendly web interface, and the entered data is processed dynamically by Java Servlets. The system automatically generates a unique order ID using customer name parts and order date through string concatenation techniques.

Problem Statement

Many businesses still use manual methods such as handwritten records, spreadsheets, and phone calls to manage customer orders. These traditional systems are time-consuming, inefficient, and prone to human errors. Problems such as duplicate records, delayed order processing, difficulty in tracking orders, and improper management of customer feedback are common in existing systems.

Objectives

1. To design a web-based order management system
2. To store customer order details in a database using JDBC
3. To generate unique order IDs dynamically
4. To analyze customer feedback strings for positive words
5. To manage orders using Java Collections

Challenges

- Ensuring uniqueness of generated order IDs
- Handling multiple customer orders simultaneously
- Maintaining secure database connectivity using JDBC

Scope

The scope of the proposed Online Order Management System is to provide a web-based platform for efficient customer order handling and management. The system allows customers to place orders easily through an online form, while businesses can securely store, retrieve, and manage order details using database technologies.

The application can be widely used in restaurants, online shopping stores, food delivery services, and small business order platforms where fast and organized order processing is required. The system helps businesses reduce manual effort, improve order accuracy, and enhance customer experience through automated order management.

CHAPTER 2

LITERATURE SURVEY & REVIEW

LITERATURE SURVEY

Web-based order management systems are widely used in modern businesses to improve the efficiency of handling customer orders. Traditional systems such as handwritten records, phone call-based ordering, and spreadsheet management are time-consuming and prone to errors. These methods create problems such as duplicate records, delayed processing, and difficulty in tracking customer orders.

With the growth of web technologies, many developers have implemented automated order management systems using Java technologies such as Servlets, JDBC, JSP, and MySQL. Java Servlets are used for processing customer requests, while JDBC helps in secure database connectivity and data storage.

1. Manual Order Management Systems

Earlier order management systems mainly depended on handwritten records, paper bills, spreadsheets, and phone call-based ordering. These methods required continuous human effort and were highly prone to errors. Maintaining records manually became difficult when the number of customer orders increased.

Studies show that manual systems often lead to:

- Data loss
- Incorrect records
- Delayed processing
- Duplicate entries
- Difficulty in tracking customer orders

These limitations created the need for automated order management systems.

2. Web-Based Ordering Systems

With advancements in web technologies, businesses started adopting web-based ordering systems. These systems allow customers to place orders online through web forms, improving

convenience and reducing manual work.

Research indicates that web-based systems provide:

- Better customer interaction
- Centralized data management
- Improved accessibility

However, some basic systems only support order submission and do not include features such as feedback analysis, dynamic order ID generation, or proper order management.

3. Database-Driven Systems

Modern order management applications use relational databases such as MySQL for storing and managing customer order details securely. JDBC is commonly used in Java applications to establish connectivity between the application and the database.

Database-driven systems improve data organization and reduce duplication problems. They also provide efficient retrieval and secure storage of customer records. Researchers emphasize that database integration is essential for reliable and scalable web applications.

Benefits include:

- Secure data storage
- Fast retrieval of records
- Better data consistency
- Reduced data duplication

Prepared Statement is also used in JDBC to prevent SQL injection attacks and improve security.

4. Java-Based Web Applications

Java technologies are widely used for developing secure, reliable, and scalable web applications. Java Servlets handle client requests and generate dynamic responses efficiently, while JDBC manages database operations.

Java Collections Framework is useful for storing and managing order records dynamically. Collections such as Array List simplify data handling, retrieval, and processing operations.

String processing techniques are commonly used for:

- Generating unique order IDs
- Validating customer inputs
- Analyzing customer feedback

Research studies show that Java-based systems are platform-independent and suitable for modern business applications.

6.Customer Feedback Analysis Systems

Customer feedback analysis is an important feature in modern business applications. Businesses collect customer comments and reviews to understand customer satisfaction levels and improve service quality.

Many systems use string processing techniques to identify positive and negative words from customer feedback. Positive words such as *good*, *excellent*, *amazing*, and *nice* help businesses evaluate customer opinions and improve their services.

Benefits include:

- Understanding customer satisfaction
- Improving service quality
- Enhancing customer experience
- Identifying customer opinions effectively

However, many traditional systems do not provide proper customer feedback analysis features.

Summary of Literature Survey

From the literature survey, it is clear that traditional manual systems are inefficient and difficult to maintain for modern business environments. Web-based automated systems provide faster processing, secure data storage, improved customer interaction, and efficient order management.

Research studies highlight the importance of Java technologies, JDBC, Java Collections Framework, and string processing techniques in developing reliable web applications. The proposed Online Order Management System combines all these features into a single platform to provide an efficient, scalable, and user-friendly solution for customer order management.

CHAPTER 3

DOMAIN ANALYSIS

3.1 Project Domain Overview

The proposed **Online Order Management System** belongs to the domain of **Web-Based Business Management Systems**. In the modern digital era, businesses require fast, secure, and efficient systems to manage customer orders, maintain transaction records, and improve customer service. Traditional manual order management methods are time-consuming, difficult to maintain, and highly prone to human errors.

The project aims to automate the complete order management process using web technologies and Java-based programming techniques. The system integrates multiple domains such as **Web Application Development, Database Management, Java Programming, Customer Feedback Analysis, Security Management, and Automation Systems**.

By combining these domains, the application provides an efficient platform for handling customer orders, maintaining records, processing data securely, and improving overall business performance. The system is suitable for restaurants, online shopping platforms, retail stores, and food delivery businesses where large numbers of customer orders are handled daily.

3.2 Order Management Domain

Order management is one of the most important domains in modern business systems. It involves receiving customer orders, processing requests, storing order details, tracking order status, and maintaining transaction records efficiently.

Traditional order management systems mostly rely on manual record keeping, which creates several problems such as:

- Difficulty in tracking customer orders
- Increased chances of data loss
- Time-consuming processing
- Duplicate records
- Human errors in calculations and management

The proposed Online Order Management System automates these activities by allowing customers to place orders online and store information directly into the database.

This domain ensures:

- Efficient order processing
- Proper tracking of customer orders
- Reduced manual effort
- Faster order handling
- Improved customer service
- Accurate record maintenance

The system is highly useful for organizations that process multiple customer orders regularly.

3.3 Web Application Development Domain

Web application development forms the core domain of the proposed system. The application is developed as a browser-based platform that users can access through the internet without requiring any additional software installation.

The web application domain focuses on:

- Designing interactive user interfaces
- Developing online order forms
- Handling client-server communication
- Processing user requests dynamically
- Displaying order confirmation messages
- Providing responsive web pages

HTML is used for creating web page structures, CSS is used for styling and improving the user interface, while JSP and Servlets are used for server-side processing and business logic implementation.

This domain helps in creating a user-friendly, responsive, and efficient online platform for customers and administrators.

3.4 Database Management Domain

Database management is an essential domain of the system because all customer and order-related information must be stored securely and efficiently.

The proposed system uses the **MySQL Database Management System** to store:

- Customer details
- Product information
- Order records
- Order IDs
- Feedback data
- Transaction details

CHAPTER 4

METHODOLOGY

The methodology chapter explains the systematic approach followed to design, develop, implement, and test the proposed Online Order Management System. A proper methodology helps in achieving accuracy, reliability, and efficiency during system development. The project follows a structured process to ensure that all functional and non-functional requirements are satisfied.

The system is developed using Java Servlets, JDBC, MySQL database, Java Collections Framework, and String Processing techniques. The methodology includes requirement analysis, system design, implementation, testing, deployment, and maintenance.

4.2 Research Methodology

The Online Order Management System follows the Waterfall Model for software development. In this model, each phase is completed before moving to the next phase. This approach helps in proper planning, documentation, and implementation of the project.

The major phases included are:

- Requirement Analysis
- System Design
- Implementation
- Testing
- Deployment
- Maintenance

This structured methodology ensures smooth development and proper integration of all modules.

4.3 Methodology Phases

4.3.1 Requirement Analysis

Requirement analysis is the first phase of the project. In this phase, the existing order management methods and their limitations were studied carefully.

Activities performed include:

- Studying manual order management systems
- Identifying problems such as duplicate records and delayed processing
- Collecting functional requirements
- Identifying security and performance requirements
- Preparing requirement specifications

This phase helped in defining the objectives and functionality of the system clearly.

4.3.2 System Design

In the system design phase, the structure and workflow of the application were planned.

Design activities include:

- Designing the system architecture
- Designing customer order forms
- Designing database tables
- Planning order ID generation logic
- Designing feedback analysis mechanism
- Defining data flow between modules

The design phase ensures that the system is scalable, secure, and easy to maintain.

4.3.3 Use Case Modeling

Use case modeling helps in understanding the interaction between users and the system.

Main Actors:

- Customer
- Administrator
- System

Major Use Cases:

- Submit order

- Generate order ID
- Store order details
- Analyze customer feedback
- Retrieve order records

Use case diagrams help in visualizing the functionality of the system.

4.3.4 System Architecture

The proposed system follows a three-tier architecture.

1. Presentation Layer

- Provides user interface for order submission
- Developed using HTML and JSP

2. Application Layer

- Handles business logic and request processing
- Implemented using Java Servlets

3. Database Layer

- Stores customer and order details securely
- Implemented using MySQL database and JDBC

This architecture ensures better performance and maintainability.

4.3.5 Implementation

In the implementation phase, the actual coding of the system was carried out based on the design specifications.

Implementation activities include:

- Developing online order form
- Creating Java Servlets for processing requests
- Implementing JDBC connectivity

- Developing order ID generation logic
- Implementing feedback analysis using string operations
- Managing records using Java Collections
- Integrating all modules into a complete application

Each module was developed and tested independently before integration.

4.3.6 Order ID Generation Method

The system automatically generates a unique order ID using customer name parts and order date.

Example:

```
String orderId = name.substring(0,3).toUpperCase() + date;
```

Generated ID Example:

PRI20260514

This method ensures uniqueness and easy order tracking.

4.3.7 Feedback Analysis Method

Customer feedback entered as strings is analyzed using string processing techniques.

Positive words checked include:

- good
- excellent
- amazing
- fast
- nice

Example:

```
if(feedback.contains("good"))
```

This helps businesses understand customer satisfaction levels.

4.3.8 Collections Usage

Java Collections Framework is used to store and manage order records dynamically.

Example:

```
ArrayList<Order> orders = new ArrayList<>();
```

Benefits include:

- Efficient data storage
- Easy retrieval of records
- Dynamic data management

4.3.9 Testing

Testing was performed to ensure proper functionality of the system.

Types of testing include:**Unit Testing**

Testing individual modules such as order submission and database connectivity.

Functional Testing

Verifying whether all system functionalities work correctly.

Integration Testing

Ensuring proper communication between Servlets, database, and frontend modules.

Error Handling Testing

Checking system behavior for invalid inputs and errors.

Testing helped identify and fix issues before deployment.

4.3.10 Deployment

After successful testing, the system was deployed in a controlled environment.

Deployment activities include:

- Configuring Apache Tomcat server
- Setting up MySQL database
- Deploying project files

- Testing application in real environment

This phase ensures the application is ready for practical usage.

4.4 Tools and Technologies Used

The following tools and technologies were used in the project:

Component	Technology Used
Frontend	HTML, CSS
Backend	Java Servlets
Database	MySQL
Connectivity	JDBC
Server	Apache Tomcat
IDE	Eclipse

These technologies provide a strong platform for developing efficient web applications.

4.5 System Workflow

The system follows a structured workflow for processing customer orders:

1. Customer submits order through web form
2. Servlet processes customer request
3. Unique order ID is generated
4. Order details are stored in database
5. Customer feedback is analyzed
6. Orders are stored using Collections
7. Dynamic confirmation message is displayed

This workflow ensures smooth and efficient order management.

CHAPTER 5

REQUIREMENT SPECIFICATION

Requirement specification defines the hardware, software, functional, and non-functional requirements needed for the development and execution of the proposed Online Order Management System. These requirements ensure that the system functions efficiently, securely, and reliably.

The requirement specification helps developers understand the resources, tools, and functionalities necessary for implementing the application successfully.

5.1 Hardware Requirements

The hardware requirements specify the minimum and recommended hardware needed to develop and run the system efficiently.

5.1.1 Minimum Hardware Requirements

- Processor: Dual-Core Processor (2.0 GHz or above)
- RAM: 4 GB
- Storage: 250 GB Hard Disk
- Display: 1280 × 720 Resolution
- Input Devices: Keyboard and Mouse
- Network: Basic Internet Connection

5.1.2 Recommended Hardware Requirements

- Processor: Intel i3 / AMD Ryzen 3 or above
- RAM: 8 GB
- Storage: 500 GB SSD/HDD
- Display: Full HD (1920 × 1080)
- Input Devices: Keyboard and Mouse

5.2 Software Requirements

Software requirements define the tools, platforms, and technologies required for system development and execution.

5.2.1 Operating System

- Windows 10 / Windows 11
- Linux (Ubuntu or equivalent)

5.2.2 Browser Support

- Google Chrome
- Mozilla Firefox
- Microsoft Edge

5.3 Functional Requirements

Functional requirements describe the core functionalities that the system must perform.

5.3.1 Order Submission Module

The system allows customers to place orders through a web-based form. Customers enter details such as name, product information, quantity, contact details, and feedback. The system validates user inputs before processing the order.

Functions

- Accept customer orders through web forms
- Validate customer inputs
- Process order details dynamically
- Store order information securely

5.4 Non-Functional Requirements

Non-functional requirements define the quality attributes of the system such as performance, reliability, usability, and security.

5.4.1 Performance Requirements

- The system should respond quickly to user requests
- Order processing should be completed within a few seconds

5.4.2 Reliability

- The system should store data securely without loss
- The application should function correctly during continuous usage

5.4.3 Performance Requirements

- The system should respond quickly to user requests
- Order processing should be completed within a few seconds

5.4.4 Reliability

- The system should store data securely without loss
- The application should function correctly during continuous usage

5.4.5 Usability

- The system should provide a simple and user-friendly interface
- Customers should easily place orders without technical knowledge

5.5 Advantages of the Proposed System

- Faster order processing
- Reduced manual effort

5.6 Summary

The requirement specification clearly defines the hardware, software, functional, and non-functional requirements needed for developing the Online Order Management System. These requirements ensure that the application performs efficiently, securely, and reliably while providing better customer service and automated order management

CHAPTER 6

RESULTS AND DISCUSSION

The Results and Discussion chapter explains the performance and output of the proposed Online Order Management System after implementation and testing. The system was tested under different conditions to verify its accuracy, efficiency, and reliability. The application successfully performs order submission, order ID generation, database storage, feedback analysis, and dynamic confirmation message display using Java technologies.

6.2 Result

The Online Order Management System successfully achieved all the project objectives. Customer orders were submitted and stored correctly in the database using JDBC connectivity. Unique order IDs were generated successfully, and customer feedback was analyzed using string processing techniques. The system efficiently managed order records using Java Collections and displayed dynamic confirmation messages after successful order placement.

The testing results showed that the application performs accurately, reduces manual effort, improves order processing speed, and provides better customer interaction.

Test Cases

Test Case ID	Description	Input Conditions	Expected Result	Observed Result	Status
TC-01	 Order Submission	Valid customer data	Order stored & ID generated	Order created successfully	✓ Pass
TC-02	 Order Submission	Missing required fields	Validation error message	Proper error displayed	✓ Pass
TC-03	 Order ID Generation	Multiple order submissions	Unique IDs generated	No duplication observed	✓ Pass
TC-04	 Database Storage	Valid order data	Data stored correctly	Data stored successfully	✓ Pass
TC-05	 Order Retrieval	Existing records	All orders retrieved	Records displayed correctly	✓ Pass
TC-06	 Feedback Analysis	Positive feedback text	Positive words identified	Feedback analyzed correctly	✓ Pass
TC-07	 Collection Management	Multiple order records	Records managed efficiently	Orders retrieved properly	✓ Pass
TC-08	 System Performance	Multiple users	Fast response (< 2 sec)	Avg. response ~ 1.5 sec	✓ Pass
TC-09	 Dynamic Message Display	Successful order placement	Confirmation message displayed	Message shown correctly	✓ Pass
TC-10	 Data Integrity	Insert → Retrieve operation	Consistent data	No mismatch observed	✓ Pass

Figure 6.1: Test Cases for Online Order Management System

6.4 Discussion

The testing results confirm that the proposed Online Order Management System performs all operations efficiently and accurately. The application successfully handles customer order submission, order ID generation, secure database storage, feedback analysis, and order management operations.

The use of Java Servlets improves server-side request processing and dynamic interaction with users. JDBC connectivity ensures secure communication between the application and the MySQL database. Java Collections Framework simplifies record management and improves data handling efficiency.

String processing techniques effectively analyze customer feedback and generate unique order IDs dynamically. The system provides fast response time and smooth interaction between different modules.

The application also reduces manual workload, minimizes errors, and improves customer experience through automated processing and dynamic confirmation messages. Overall, the developed system provides a reliable, secure, and scalable solution for modern online order management requirements.

6.5 Implementation Details

6.5.1 Frontend Implementation

The frontend of the application was developed using:

- HTML
- CSS
- JSP

The frontend provides:

- Customer order form
- User-friendly interface
- Dynamic confirmation messages

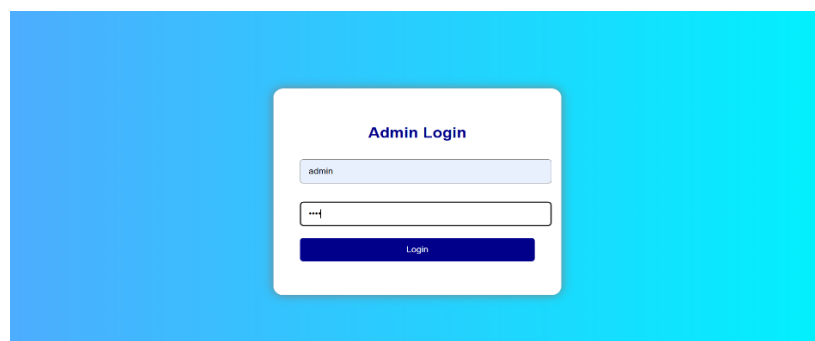


Fig 6.5.1: Admin Login Page

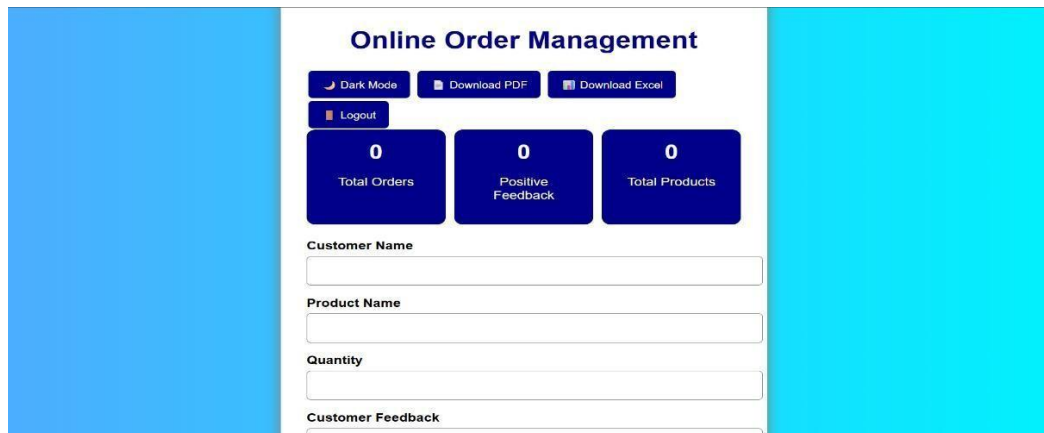


Fig 6.5.1: Online Order Management Interface

6.5.2 Backend Implementation

The backend was implemented using Java Servlets.

Functions performed include:

- Processing customer requests
- Generating order IDs
- Handling feedback analysis
- Managing business logic

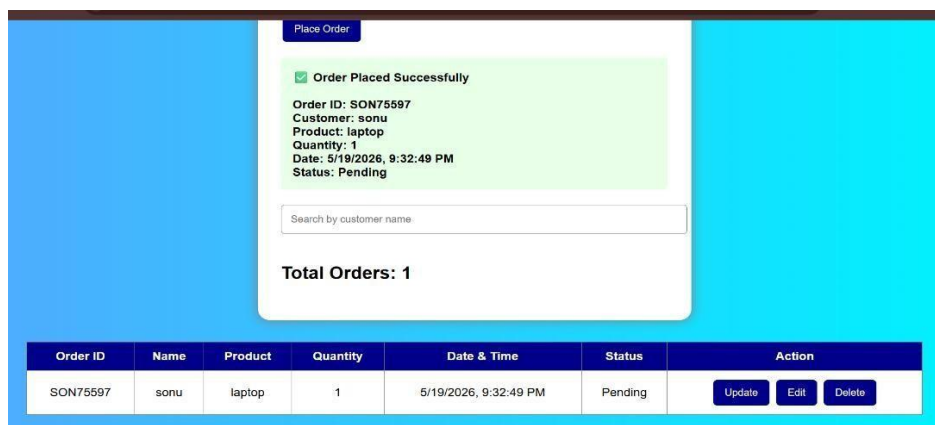


Figure 6.5.2: Successful Order Submission

6.5.3 Database Implementation

MySQL database was used for storing:

- Customer details
- Product details
- Order records
- Feedback information
- Generated order IDs

JDBC was used for database connectivity and secure data operations.

AutoSave

OFF

orders-report (1) - Protected... - Saved to this PC

Search

FileHomeInsertDrawPage LayoutFormulasDataReviewViewHelp

PROTECTED VIEWBe careful—files from the Internet can contain viruses. Unless you need to edit, it's safer to stay in Protected View.

Enable Editing

CommentsShare

A B C D E F G H I J K L M N O P Q R S T U

1nameproductquantityfeedbackorderiddatestatus

2summu1good51JMS4845/19/2026Pending

3veena1goodVh2-28D-45/19/2026Pending

4ramya1badRAM663415/19/2026Pending

5sonu1very good5ON62625/19/2026Pending

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Ready

Orders

Fig 6.5.3: Customer Order Records Stored in Database

6.5.4 Collections Implementation

Java Collections such as ArrayList were used to:

- Store order records dynamically
- Retrieve records efficiently
- Manage multiple customer orders

Example:

```
ArrayList<Order> orders = new ArrayList<>();
```

6.5.5 Feedback Analysis Implementation

Customer feedback strings were analyzed using string operations.

Positive words checked include:

- good
- excellent
- amazing
- fast
- nice

6.6 Advantages of the System

- Faster order processing
- Reduced manual effort
- Secure database storage
- Efficient order management
- Dynamic confirmation messages
- Improved customer interaction

6.7 Limitations

- No online payment integration
- No admin login module
- Limited feedback sentiment analysis
- Mobile application support not included

6.8 Future Enhancements

The system can be improved in the future by adding:

- Online payment gateway integration
- Email and SMS notifications
- Admin dashboard for monitoring orders
- Advanced feedback sentiment analysis using AI
- Mobile application support

6.9 Summary

The Results and Discussion chapter confirms that the Online Order Management System successfully fulfills all functional requirements. The system provides efficient customer order handling, secure data storage, dynamic processing, and improved customer interaction using Java web technologies. Overall, the application offers a reliable and scalable solution for automated online order management.

CONCLUSION AND FUTURE SCOPE

Conclusion

The **Online Order Management System** successfully demonstrates how Java web technologies can be used to build an efficient, reliable, and user-friendly order processing application. The system automates the complete process of customer order submission, order ID generation, database storage, feedback analysis, and order management, thereby reducing manual effort and improving operational efficiency.

By providing a structured web-based interface, the system allows customers to place orders easily, while administrators can manage and monitor order records effectively. The automatic generation of unique order IDs ensures proper order tracking and avoids duplication of records. The use of JDBC for database connectivity guarantees secure and accurate data storage, while Java Collections Framework enables efficient management and retrieval of customer order records.

Future Scope

Although the system successfully fulfills the basic requirements of online order management, several enhancements can be implemented in the future to improve functionality, usability, and scalability.

- **Online Payment Gateway Integration**

Integrate secure online payment systems such as Google Pay, PhonePe, Paytm, or credit/debit card payment facilities for real-time transactions.

- **Email and SMS Notification System**

Automatically send order confirmation messages, delivery updates, and notifications to customers through email and SMS services.

- **User Authentication and Role Management**

Implement secure login functionality with different user roles such as Admin, Staff, and Customer to control system access and permissions.

- **Admin Dashboard**

feedback analysis, and system activities.

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